

DSA Roadmap to Crack Companies like Zoho (Beginner → Intermediate)

Phase 1: Beginner Level – Build Logic Foundation

Goal: Understand how to think and write logic.

Topics:

- Time & Space Complexity
- Arrays
- Strings

Problem Questions:

Arrays

1. Find the largest element in an array
2. Find the second largest element
3. Reverse an array
4. Check if array is sorted
5. Move all zeros to end

Strings

6. Check if a string is palindrome
7. Reverse a string
8. Count vowels and consonants
9. Find frequency of characters
10. Remove duplicates from string

Outcome:

You will learn basic thinking and logic building.

Timeline: 2 weeks

Phase 2: Beginner → Intermediate – Pattern Building

Goal: Recognize interview patterns.

Topics:

- Hashing
- Two Pointer
- Sliding Window

Problem Questions:

Hashing

11. Find duplicate element in array
12. Find first non-repeating character
13. Count frequency of each element
14. Two sum problem

Two Pointer

15. Reverse array using two pointer
16. Check palindrome using two pointer
17. Pair with given sum in sorted array

Sliding Window

18. Maximum sum subarray of size k
19. Longest substring without repeating characters
20. Find all subarrays of size k

Outcome:

You start recognizing patterns.

Timeline: 3–4 weeks

Phase 3: Intermediate Level – Interview Core Topics

Goal: Think like interviewer.

Topics:

- Binary Search
- Recursion
- Sorting

Problem Questions:

Binary Search

21. Binary search in sorted array
22. First and last occurrence
23. Find square root using binary search

Recursion

24. Factorial using recursion
25. Fibonacci using recursion
26. Reverse array using recursion

Sorting

- 27. Bubble sort
- 28. Selection sort
- 29. Insertion sort

Outcome:

Strong problem solving ability.

Timeline: 3–4 weeks

Phase 4: Most Important – Optimization + Dry Run

This is the exact step that helped me crack Zoho.

Practice problems like:

- 30. Two sum optimized
- 31. Find duplicate using hashing
- 32. Longest substring optimized
- 33. Maximum subarray sum (Kadane's Algorithm)

Practice rules:

- Write code on paper
- Dry run manually
- Optimize time complexity

This builds interview thinking.

Daily Practice Plan (1–2 hours per day)

Daily:

- 1 Easy problem
- 1 Medium problem
- Dry run on paper

Consistency > Quantity

Real Interview Truth

Companies like Zoho expect:

- Strong basics
- Clear thinking
- Ability to optimize
- Confidence in explaining

Not 500 problems. Even 150–200 quality problems is enough.

Exact Order to Follow

1. Arrays
 2. Strings
 3. Hashing
 4. Two Pointer
 5. Sliding Window
 6. Binary Search
 7. Recursion
 8. Sorting
 9. Optimization
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Golden Rule

Don't memorize solutions.

Understand patterns.

DSA is not about syntax.

DSA is about thinking.